1. Consider a classroom that is roughly 5 m × 20 m × 3 m. on a hot and humid day, the air conditioner suddenly stops working at the moment when the temperature and the pressure of the room are 20 °C and 1 bar, respectively. There are 100 people in the class, each losing energy to the room at the average rate of 150 W. Assume that the walls, ceiling, floor, and furniture are perfectly insulating and do not absorb any heat. How long will the physical chemistry class last if the professor has agreed to dismiss the class when the air temperature in the room reaches body temperature, 37 °C. Take $C_{p,m} = 7R/2$ for air and assume that the loss of air to the outside as the temperature rises may be neglected.

2. Ethanol, C_2H_5OH, is brought to the boil at 1 bar pressure. When an electric current of 0.682 A from a 12.0 V supply is passed for 500 s through a heating coil immersed in the boiling liquid, it is found that the temperature remains constant but 4.33 g of ethanol is vaporized. What is the standard enthalpy of vaporization of ethanol at its boiling point?

3. The standard reaction enthalpy ($\Delta_r H^\ominus$) for the hydrogenation of propene $C_3H_6(g) + H_2(g) \longrightarrow C_3H_8(g)$ is -124 kJ·mol^{-1} and the $\Delta_r H^\ominus$ for the combustion of propane $C_3H_8(g) + 5\ O_2(g) \longrightarrow 3\ CO_2(g) + 4\ H_2O(l)$ is -2220 kJ·mol^{-1} at room temperature. Calculate $\Delta_r H^\ominus$ for the combustion of propene at this temperature. [*Hint:* $\Delta_r H^\ominus$ for the dissociation of water $H_2O(l) \longrightarrow H_2(g) + (1/2)\ O_2$ is +286 kJ·mol^{-1}].

4. The standard enthalpy of formation of gaseous H_2O at 25 °C is $\Delta_f H^\ominus = -241.82$ kJ·mol^{-1}. Estimate its value at 100°C given the following values of the standard molar heat capacity at constant pressure ($C^\ominus_{p,m}$/ J K^{-1} mol^{-1}): H_2O(g): 33.58, H_2(g): 28.44, O_2(g): 29.37. (Assume that the heat capacities are constant in this temperature range).

5. (i) Liquid helium boils at 4 K while liquid hydrogen boils at 20 K at 1 bar. What is the efficiency of a reversible (Carnot) heat engine operating between two heat reservoirs at these temperatures. (ii) If we wanted the same efficiency as above for a reversible heat engine with a cold reservoir at ordinary temperature (25 °C), what must the temperature of the hot reservoir be?

6. The efficiency of a heat engine that operates between two heat reservoirs at 100 °C and 0 °C is 75% of the value for a Carnot engine. If this heat engine absorbs 10 kJ of heat from the hot reservoir, then in each cycle (i) how much work is done and (ii) how much heat is discharged to the cold reservoir?

7. Draw the schematic diagrams for the Carnot cycle showing (i) change in entropy with temperature, and (ii) change in entropy with volume.

8. (a) The internal pressure of a gas $\pi_T = (\partial U/\partial V)_T$ is a measure of the strength of the cohesive forces in the sample. What is the value of π_T for an ideal gas? Explain your answer in the light of Joule's experiment.
   (b) The expansion coefficient $\alpha = (1/V)(\partial V/\partial T)_p$ is a measure of how easily a sample can be expanded by increasing the temperature at a constant pressure. What is α for an ideal gas?
   (c) Isothermal compressibility $\kappa_T = -(1/V)(\partial V/\partial p)_T$ is a measure of how easily a sample can be compressed by increasing the pressure at a constant temperature. What is κ_T for an ideal gas?
   (d) Show that $(\partial U/\partial T)_p = \alpha\pi_T V + C_V$.
   (e) Show that $(\partial H/\partial T)_V = (1 - \alpha\mu/\kappa_T)C_p$ where $\mu = (\partial T/\partial p)_H$ is a consant ("Joule-Thomson coefficient") that indicates whether a sudden lowering of pressure of a real gas under adiabatic conditions can lead to cooling or heating of the gas.
   (f) Show that $C_p - C_V = \alpha(p + \pi_T)V$ for the general case.
   (g) Using the results from (a) and (b), show that the above relation reduces to $C_p - C_V = nR$ for an ideal gas.